Dantrolene *Prodigy Anesthesia*

Dantrolene is a spasmolytic agent that is chemically similar to phenytoin that is used to treat malignant hyperthermia. Patients with malignant hyperthermia have an alteration in calcium induced calcium release via the ryanodine channel or an inability of the sarcoplasmic reticulum to sequester calcium via the calcium transporter. After the patient with malignant hyperthermia is exposed to a triggering agent, there is an abrupt and prolonged released of calcium which results in widespread muscle contraction, increased lactic acid production and buildup, and increased temperature.

Dantrolene binds to the ryanodine-1 channel and inhibits the calcium channel in the sarcoplasmic reticulum. Cardiac muscle tissue and smooth muscle tissue are not affected greatly by this effect because they employ the slightly different ryanodine-2 channel.

Treatment of malignant hyperthermia with dantrolene is usually initiated with a dose of 2.5 mg/kg IV then repeating doses of 1 mg/kg to control symptoms up to a maximum dose of 10 mg/kg.

Katzung BG, Masters, SB, & Trevor AJ. Basic and Clinical Pharmacology. 11th ed. New York: McGraw-Hill; 2009: 465.

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